

ALAN HUYNH

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SKILLS

Languages: Python, SQL, JavaScript, C/C++, Java

Machine Learning: PyTorch, scikit-learn, TensorFlow, computer vision, time-series forecasting, PINNs

Data & Geospatial: NumPy, Pandas, PySpark, Rasterio, GeoPandas, GDAL, Sentinel-2, COG/STAC

Cloud & MLOps: AWS, MLflow, TensorRT, Docker, Terraform, Git, Linux

Domain: AEMO NEM, Battery, Emissions & Environmental modelling, Remote Sensing, Edge Inference, DevOps

EXPERIENCE

Machine Learning Consultant

Jun. 2025 – Present

ProtoFlow Pty Ltd

Melbourne, VIC

- Built an AI subtitling platform that processed 1,000+ hours of video across 10 languages. Multi-agent MQM quality pipeline over Claude, Gemini and GPT using Deepgram ASR with human-in-the-loop review.
- Authored 20+ problems for frontier LLM training, each required to defeat most current models.
- Built an automated AEMO NEM trading system for a Sydney energy startup using mathematical optimisation for dispatch scheduling. 29% revenue uplift over the production model.
- Led design and implementation of a proprietary POS and kiosk platform for a multi-site F&B operator serving 1000+ customer on average per day. Formed and led a three-person operations team from client staff, owning delivery from scoping to handover.
- Built a Moodle LLM assistant for an education provider, 200 students daily.

Machine Learning Engineer

Dec. 2024 – Feb. 2025

Fair Dinkum Systems

Melbourne, VIC

- Dual-stream Vision Transformers in PyTorch for battery remaining-useful-life prediction, evaluated across 6 datasets including the Stanford fast-charging dataset.
- Time-series feature pipeline: wavelet transforms, spectrograms, Fourier analysis.
- Contrastive learning for out-of-domain robustness. 4-person team; pytest and GitHub Actions.

Software Engineer

Sep. 2024 – Sep. 2025

Math Song Pty Ltd

Melbourne, VIC

- Led full-stack Django platform development, automating manual business operations.
- Payment integrations handling \$200K+ per month in transaction volume, no payment-path incidents during tenure
- GCP infrastructure: Docker, Terraform, CI/CD pipelines, VM provisioning.

Machine Learning Engineer

Feb. 2024 – Sep. 2024

AIgorithm

Melbourne, VIC

- Delivered satellite-derived emissions metrics at national ABS Mesh Block resolution (368k units) for a carbon initiative serving banking and retail clients.
- Semi-supervised computer vision on multi-spectral imagery; self-supervised pretraining to adapt European land-classification frameworks to Australian conditions.
- LSTM to XGBoost pipeline estimating net carbon exchange and methane from NASA Earthdata inputs over ALUM land-use classes; containerised to AWS ECR and Fargate.
- Production ETL for multi-spectral imagery on AWS Glue (Spark), EMR and Lambda.

Data Scientist

Jan. 2022 – Jan. 2024

Lux Aerobot

Melbourne, VIC

- Computer vision for wildfire detection from multi-spectral imagery on stratospheric balloon platforms, supporting Canadian and Australian environmental monitoring and disaster recovery.
- Optimised models for NVIDIA Jetson quantisation and TensorRT, cutting model size 60% for real-time field inference.
- Designed the full gondola stack (hardware, firmware, deployment software) and AWS training pipelines on SageMaker, Neo and Greengrass.

SPEAKING

Physics-Informed Neural Networks: Less Data, More Physics <i>DDD Melbourne</i>	Sep. 2025
No ML is (Probably) the Best ML <i>DDD Melbourne</i>	Dec. 2024
No ML is probably the best ML <i>AICamp Melbourne</i>	Nov. 2024

PROJECTS

Green Battery Hack: First Place, Innovation Track <i>MLAI Australia</i>	Apr. – May 2024 <i>Melbourne, VIC</i>
- Engineered ML model for AEMO spot market trading using simulation and reinforcement learning-inspired strategies. - Implemented a custom loss function that quantified the difference between realized profits and maximum achievable profits, driving the model to learn optimal trading decisions. - Experimented with Q-Learning variations, time series, indicator-based models (EMA), and Optuna optimization to enhance trading strategies.	
Airborne Microplastics: Physics-Informed Neural Network <i>PyTorch Lightning</i>	Apr. 2025 – Sep. 2025
- Physics-informed multi-head CNN estimating 3D microplastic mass mixing ratios and 2D deposition from MOZART-4 output. 6 particle sizes, 48 altitude levels, 0.1 degree global. - Controlled ablation on the advection-diffusion-settling PDE loss: R^2 0.61 best configuration against a 0.33 baseline. Mass-conservation constraint implemented, evaluated, rejected.	
Torch Activations <i>PyTorch, Poetry</i>	Apr. 2023 – Present
- Developed and published a PyPI library, 'Torch Activations,' implementing 430 activation functions for PyTorch from mathematical representations. - Ensured 100% test coverage and implemented robust CI/CD pipelines using GitHub Actions and Poetry for automated deployment and continuous integration. - Established and managed a structured open-source project, including detailed issue tracking, pull request guidelines, and contribution documentation on ReadTheDocs.	
Melbourne Rooftop Solar Potential <i>laspy, rasterio, GeoPandas, Shapely, pvlib</i>	Nov. 2025
- Analysis with Melbourne aerial point cloud using SMRF ground classification, building segmentation, RANSAC roof plane extraction, per-plane tilt and azimuth across 26,519 buildings. - Ray-cast horizon profiles for shading with hourly irradiance modelling over a full year and sensitivity analysis across packing factor and performance ratio.	

EDUCATION & CERTIFICATION

AWS Certified Machine Learning Specialty <i>MLS-C01</i>	Mar. 2025 <i>Expires Mar. 2028</i>
Melbourne Polytechnic <i>Bachelor of Information Technology</i>	Feb. 2024 – Oct. 2026 <i>Melbourne, VIC</i>
Deakin University <i>Accelerated Program</i>	Feb. 2023 – Oct. 2023 <i>Melbourne, VIC</i>

OTHER

Writer on environment and AI.
Mentor at CISSA, University of Melbourne (2025).
Founder of Arcaea Melbourne community meetups.
Ultra-endurance gravel cyclist with 300+ km single-day efforts.